

Here are your notes for "Lowess and Loess, Clearly Explained!!!".

# Lecture Notes: Lowess and Loess (Fitting a Curve to Data)

## 1. The Goal: Beyond a Straight Line

- **Linear Regression** (Least Squares) fits a single **straight line** to all the data [00:45].
- **Lowess/Loess** (Locally Weighted Scatterplot Smoothing) is a more flexible method that fits a **smooth curve** to the data. It's designed to capture changing trends that a single straight line would miss [01:03].

## 2. The Main Ideas

1. **Sliding Window:** Instead of looking at all the data at once, the method uses a "sliding window" to look at **small local subsets** of the data (or "blobs") [01:12].
2. **Local Fits:** Within each small window, it fits a simple line using **weighted least squares** [01:21]. The collection of all these tiny local fits creates the final smooth curve.

## 3. The Loess/Lowess Algorithm (Step-by-Step)

The algorithm repeats a process for *every single data point*:

1. **Pick a Focal Point:** Start by making the first data point the "focal point" [01:46].
2. **Define the Window:** Create a "window" that contains a set number of its *nearest neighbors* (e.g., the 5 closest points).
  - **Important:** "Closeness" is determined **only by the X-axis distance**, not the Y-axis [05:02].
3. **Assign Weights (X-axis):** Perform a **weighted least squares fit** for the points in the window.
  - The **focal point** gets the **most weight** [02:32].
  - Points farther away from the focal point (on the X-axis) get progressively **less weight** [02:40].
  - This ensures the focal point has the most influence on this specific local fit.
4. **Find the New Point:** The fitted line gives a *new* Y-value for the focal point's X-value. This new Y-value is the **first draft** of the final smooth curve [02:59].
5. **Repeat for All Points:** Slide the window to the next data point, make *it* the focal point, and repeat steps 2-4. This is done for all points, creating a full "first draft" of the curve [05:37].

## 4. Refining the Curve (Handling Outliers)

The "first draft" curve can still be pulled around by outliers [05:54]. The algorithm refines itself to fix this:

1. **Assign New Weights (Y-axis):** Go back and look at the *original* data. Calculate the vertical distance (residual) from each original point to the "first draft" curve.
2. **Identify Outliers:** Points that are *far away* from the first draft curve (like the outlier) get a **low weight**. Points that are *close* get a **high weight** [06:05].
3. **Repeat the Entire Process:** The entire sliding window process (Steps 1-5) is done over again. But this time, the weighted least squares calculation uses *both* sets of weights:
  - **Weight 1 (X-axis):** Based on local distance.
  - **Weight 2 (Y-axis):** Based on outlier-ness.
4. This refinement step can be repeated multiple times, with each iteration becoming less sensitive to the outliers and resulting in a smoother, more robust curve [07:03].

## 5. Important Options and Considerations

- **Line or Parabola?**
  - At each step, you can fit a simple **line (linear)** or a **parabola (quadratic)** within the window [07:30].
  - Fitting a parabola can sometimes capture curves more accurately [07:55].
  - In R, `lowest()` only fits a line. `loess()` can fit either (it defaults to a parabola) [08:01].
- **Window Size (Span):**
  - You must choose the size of the window, usually as a **proportion of the total data** (e.g., 1/5, 1/3, or 0.75) [08:54].
  - A **small window** (e.g., 1/5) creates a *wiggly, flexible* curve that follows local trends.
  - A **large window** (e.g., 0.75) creates a *smoother, less flexible* curve, closer to a single straight line [09:03].
- **Weight Functions:** The specific mathematical formulas used to assign weights (both X-axis and Y-axis) are somewhat arbitrary; they were chosen because they work well in practice [09:23].

[http://googleusercontent.com/youtube\\_content/13](http://googleusercontent.com/youtube_content/13)